

CASE STUDY · GOVERNMENT VERTICAL · EXPERIMENTATION

Turning Portal Decisions Into Experiments

An end-to-end A/B testing program on the Medicaid verification portal that raised completion rates, lowered cost per verification, and changed how decisions get made.

+4–6%

COMPLETION RATE LIFT

–3–5%

COST PER VERIFICATION

15+

STATE CLIENTS ROLLED OUT

500K+

TXNS POWERING DESIGN

Decisions about portal UX and pricing were running on opinion.

THE PORTAL

States use the Medicaid verification portal to confirm benefit eligibility. My team's job in the Government vertical was to grow and retain those accounts by lifting completion and cost efficiency.

WHAT WE WERE SEEING

Many sessions were starting but not completing, and several states were sensitive to cost per verification. UX, copy, and pricing varied by client — and no one could say what was actually working.



Sessions started, not finished

A meaningful share of verification sessions abandoned mid-flow — every drop-off was a cost incurred without a payoff.



Cost-per-verification scrutiny

Some states pushed back on rising verification costs — a competitive risk that contract renewals would amplify.



"What does the data say?"

Pricing and messaging debates lived in meetings. There was no defensible signal to settle them with.

Build a structured A/B testing program — not a one-off test.

THE PROPOSAL

Stop debating pricing and messaging in meetings. Stand up a repeatable A/B testing pattern so portal changes default to experiments — not opinions.



PRIMARY METRIC

Verification completion rate — completed verifications divided by started sessions.

Aligned with Sales & Product



GUARDRAILS

Total completed volume per state, latency, error rates — no win at the cost of breaking the system.

Negative-result discipline



SECONDARY METRICS

Cost per verification, revenue per verification, abandonment rate by funnel step.

Business-model aware



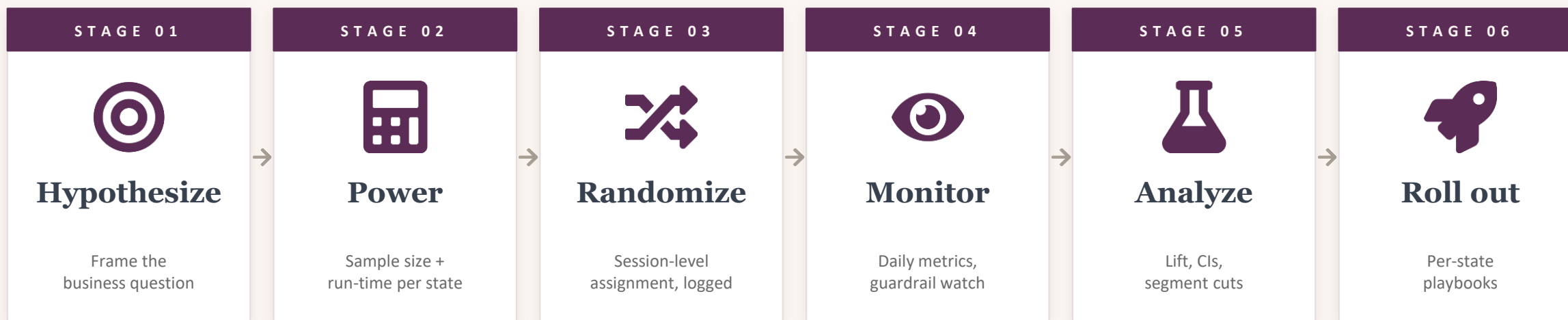
STAKEHOLDERS

Product (UX), Engineering (instrumentation), Sales (state context), legal (contract constraints).

Cross-functional from day one

Six stages — from hypothesis to playbook.

I owned every step of the experimentation lifecycle, and worked with Product, Engineering, and Sales to make the results actually change behavior.



WHY "PROGRAM" NOT "TEST"

One test answers one question. A program changes how the org makes decisions — that was the actual deliverable.

Two business goals, three metric layers, one testable hypothesis.

THE HYPOTHESIS

BUSINESS HYPOTHESIS

Clearer messaging about value and a more intuitive pricing presentation will lift completion rates and shift users toward more cost-efficient options.

STATISTICAL FORM

H_0 : $CR_A = CR_B$ (no difference)

H_1 : $CR_B > CR_A$ (treatment lifts completion)

METRIC LAYERS

PRIMARY

Completion rate

Completed verifications $\div$ started sessions. The number we're trying to move.

SECONDARY

Cost & revenue per verification

Captures whether wins on completion translate to real economics — not just clicks.

GUARDRAIL

Volume, latency, error rates

Negative trip-wires. We will not ship a winner that breaks total volume or system health.

PRINCIPLE · Pick metrics that mirror the business model. For state contracts, cost per verification matters more than clicks-per-page.

Three design choices that determined whether the test would even mean anything.

An A/B test only earns the right to a conclusion if it's designed for one. These were the calls that made the analysis defensible.



Unit of randomization

WHAT WE DECIDED

- Session-level within each state
- Stable assignment per session
- User doesn't bounce A → B mid-flow

WHY: Matched how the user experienced the change — and avoided contamination across treatments.



Per-state vs. pooled

WHAT WE DECIDED

- Per-state for high-volume clients
- Pooled with state covariate for smaller ones
- Decision driven by sample size, not preference

WHY: Some states had the volume to power a stand-alone test. Others were pooled with state included as a covariate to retain power.



Variant content

WHAT WE DECIDED

- Pricing display: per-txn vs. bundled
- Copy explaining cost differences
- Value/data-freshness messaging

WHY: Co-designed with Product and reviewed against contract language so variants were on-brand and contract-safe.

Powered to detect what would actually matter — not just what's easy.

POWER CALCULATION INPUTS

INPUT	VALUE	BASIS
Baseline completion rate	estimated from history	from 500K+ txns
Minimum detectable effect	3–4% relative lift	what's worth shipping
Significance level (α)	0.05	two-sided
Power ($1 - \beta$)	0.80 – 0.90	comfortable detection
Variance	binomial	completion is binary

Plugged into a power calculator → required N per variant → translated into run-time per state using historical daily volumes.

FROM SAMPLE SIZE TO RUN-TIME



High-volume states

Hit required sample size in 2–3 weeks — could run stand-alone tests with strong power.



Mid-volume states

Needed longer runs or pooled designs to reach detectable-effect thresholds without bleeding seasonality.



Pre-registered duration

Lock end date upfront. No early-stopping for positive results — that's how teams learn to overfit to noise.



Stop conditions

Exception: severe negative impact on guardrails would trip the kill switch and end the test early.

PRINCIPLE · Underpowered tests are worse than no test — they manufacture noise and call it signal.

QA before traffic, dashboards during traffic.

IMPLEMENTATION

1

Wire experiment flags

Worked with Engineering to add server-side flags so the portal could render Variant A or B per session.

2

Verify event logging

Confirmed session start, step views, completion, and cost events all flowed cleanly into BigQuery with experiment ID + variant tagged.

3

QA assignment proportions

Ran in test environment, then live with a small fraction of traffic to verify 50/50 split, no skew across covariates.

DAILY MONITORING (TABLEAU)



Daily completion rate by variant

Watched for divergence — and for unexpected mid-test reversals that could signal a bug.



Daily cost per verification

Secondary metric tracked throughout, not just at the end. Surprises in either direction get caught early.



Volume balance check

Confirmed 50/50 split held across days, states, and times — drift here invalidates the analysis.



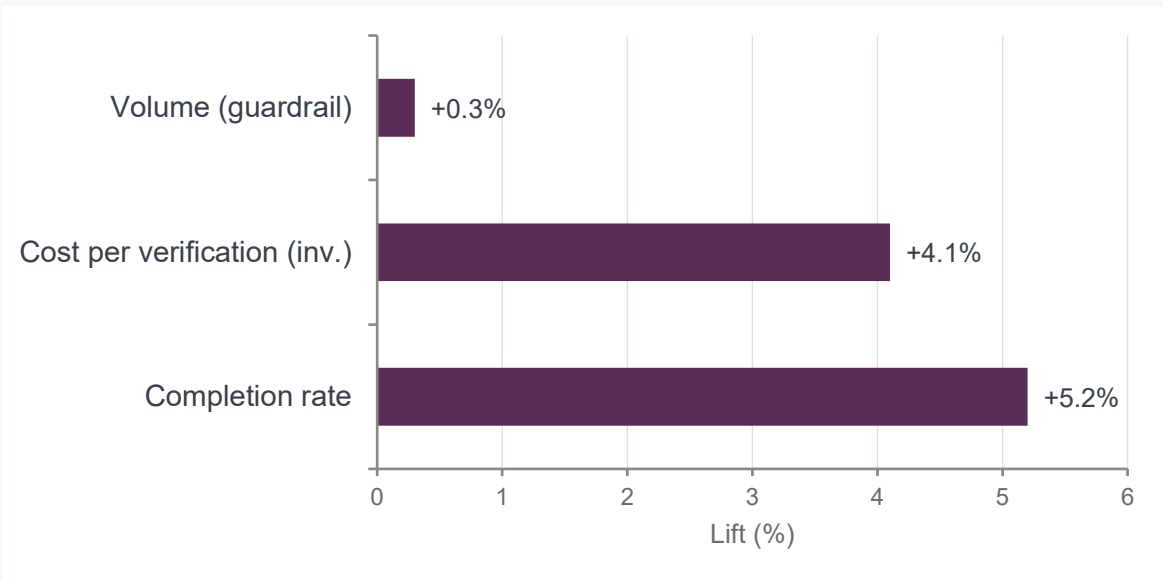
Guardrail trip-wires

Latency, error rates, total completed volume — alerted if any crossed pre-defined negative thresholds.

GUARDRAIL · Monitoring catches bugs and trip-wires — not significance. Final reads happen at pre-registered end dates.

Lift, intervals, segment cuts — not just a single p-value.

ESTIMATED LIFT BY METRIC (95% CI)



Illustrative — point estimates with 95% CIs. Volume guardrail flat (within noise), as required.

ANALYSIS DISCIPLINE



Two-sample tests on proportions

Standard for completion-rate comparisons. Logistic regression with state covariate when pooling.



Confidence intervals, not just p-values

A p-value alone hides effect size. Stakeholders need to see how big the lift actually is.



Heterogeneity by segment

Sliced by state, time of day, user type. A pooled win that hides per-state losses is not actually a win.



No peeking, no p-hacking

Pre-registered analysis. Final read at the planned end date — discipline that earns the right to act.

OUTCOME · Variant B lifted completion 4–6%, lowered cost per verification 3–5% — with no negative impact on overall volume.

From winning variant to standard practice across the vertical.

A test isn't done at the analysis. It's done when the winning variant is in production at scale and the next decision defaults to an experiment.

1 Per-state recommendations

Presented to Sales and Product as actionable: which states roll out as-is, which need contract-language tweaks, and which need a follow-up test.

2 15+ state rollout

Promoted Variant B to 100% traffic at every eligible state, retired the experiment flag, and logged the change in a central experiments registry.

3 Experimentation playbook

Documented the methodology so the next portal change defaults to an A/B test instead of a one-off launch — the actual durable deliverable.

Both metrics moved — and the org's decision-making did too.

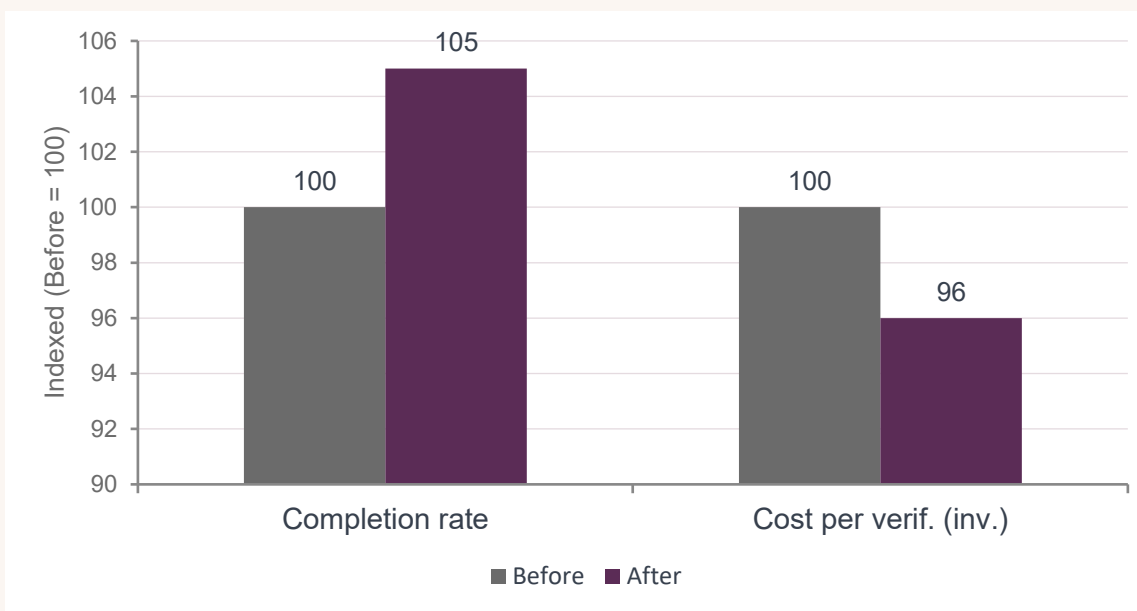
HOW WE MEASURED

Compared in-test treatment vs. control, then verified the gains held in production after rollout — with seasonality and policy-change adjustments.

WHAT MOVED

- Completion rate — primary outcome
- Cost per verification — secondary outcome
- Volume (guardrail) — held flat
- Decision-making culture — "what does the test say?"

BEFORE vs. AFTER ROLLOUT



Cost per verification shown as inverse so direction is consistent (higher = better).



HEADLINE IMPACT

+4 to +6% completion • **–3 to –5% cost per verification** across 15+ state clients, with overall volume held flat.

The scoreboard.



COMPLETION RATE LIFT

+4 to +6%

primary outcome, in-test and post-rollout



COST PER VERIFICATION

-3 to -5%

secondary outcome — economic, not just behavioral



STATE CLIENTS ROLLED OUT

15+

winning variant promoted across the vertical



DURABLE DELIVERABLE

Playbook

next portal change defaults to an experiment

QUALITATIVE WINS

Decision-making shifted from opinion to evidence · Sales gained per-state recommendations they could act on · Methodology now applied to other portal changes across the vertical.

What worked, what I'd extend next.

WHAT WORKED



Building a program, not a test

The playbook and registry outlasted any single experiment — the durable lift came from changing how decisions get made.



Pre-registering analysis

No peeking, no p-hacking. The discipline made the wins defensible and gave Sales something they could carry into client conversations.



Guardrails that could kill a winner

Refusing to ship a primary-metric win if guardrails moved in the wrong direction — that's what made the program credible long-term.

WHAT I'D EXTEND NEXT



Bayesian methods for small states

For low-volume states, frequentist power requirements drag tests for months. Bayesian approaches with informative priors would let us decide faster.



Multi-armed bandit for low-stakes

Reserve the full A/B program for high-stakes UX changes. For minor copy tweaks, bandits can balance learning and traffic optimization automatically.



Cross-test meta-analysis

After a year of experiments, mine the registry for patterns — which message types tend to win, which states differ most. Compounds the playbook value.

From opinion to evidence — as the default mode of operating.

01

The win is the program, not the test.

One A/B answers one question. A repeatable program changes the operating culture — and that's what compounds.

02

Power is a discipline, not a calculator step.

Underpowered tests manufacture noise. Pre-registering effect size, alpha, and run-time is what earns the right to act on the result.

03

Guardrails matter as much as winners.

Refusing to ship a primary-metric win that breaks volume or latency is what makes a program trustworthy over time.

Thank you · Questions?